

A Control-Theoretic View of Mamba on Stability and Robustness

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Abstract

Selective State Space Models (SSMs) such as Mamba have emerged as efficient alternatives to Transformers, achieving linear complexity through input-dependent parameterization. However, this selectivity transforms the system from linear time-invariant (LTI) to linear parameter-varying (LPV), where individually stable matrices can produce unbounded trajectories under switching. Existing work focuses on empirical performance, leaving global stability, robustness bounds, and practical certification unresolved. This paper develops a control-theoretic framework providing a comprehensive stability and robustness analysis for selective SSMs. We prove BIBO stability by viewing selective scans as continuous-time LTI sampling and establish two-term robustness bounds with linear growth in sequence length. For general LPV systems, we provide common quadratic Lyapunov function conditions and develop algorithms to extract certificate constants directly from network weights. These results bridge control theory and SSM architectures, providing verifiable certificates for individual selective-SSM layers as a step toward formal guarantees for deployment.

1. Introduction

Sequence modeling lies at the heart of modern machine learning, with applications spanning natural language processing, computer vision, speech recognition, and time-series analysis. While Transformers (Vaswani et al., 2017) have dominated this landscape, their quadratic complexity in sequence length poses fundamental scalability challenges

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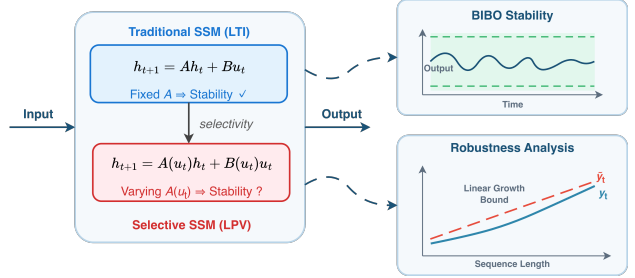


Figure 1. Control-theoretic analysis of selective SSMs

for long-context applications. This limitation has driven intense research into efficient alternatives that maintain modeling capacity while achieving sub-quadratic complexity.

State Space Models have emerged as a principled approach to this challenge, drawing on decades of control theory to enable efficient sequence processing (Gu et al., 2022). The key insight is that linear dynamical systems admit both recurrent and convolutional views, enabling hardware-efficient implementations via parallel scans (Smith et al., 2023). The HiPPO framework (Gu et al., 2020) provided theoretical grounding by connecting SSM initializations to optimal polynomial projections of signal history, yielding interpretable memory mechanisms.

A major breakthrough came with Mamba (Gu & Dao, 2024), which introduced selective state spaces where SSM parameters depend on the input content. This modification enables the model to dynamically adjust its information routing based on context, analogous to the selective attention in Transformers but with linear complexity. Mamba and its variants have achieved empirical success across modalities (Dao & Gu, 2024; Zhu et al., 2024).

However, this selectivity fundamentally changes the system’s mathematical structure. Traditional SSMs are Linear Time-Invariant (LTI) systems, whose stability is determined by spectral properties of a fixed transition matrix. Selective SSMs, in contrast, are Linear Parameter-Varying (LPV) or switched systems, where the transition matrix changes at each step based on input. A critical result from control theory is that a collection of individually stable matrices can produce unstable behavior under arbitrary switching (Liberzon, 2003). This raises fundamental questions: Under what conditions are selective SSMs globally stable? How ro-

bust are outputs to input perturbations? Can we provide verifiable certificates from network weights?

This paper develops a control-theoretic framework that provides verifiable stability and robustness guarantees for selective SSMs (as shown in Figure 1). Our key insight is that the fixed- A exponential scan $A_t = e^{\Delta_t A}$ can be viewed as exact sampling of a continuous-time LTI system, which transforms a challenging switched-system analysis into tractable continuous-time bounds. This perspective yields three main results:

- **BIBO Stability:** The standard Mamba parameterization is provably stable under mild assumptions on the Hurwitz matrix A .
- **Linear-Growth Robustness:** Output sensitivity decomposes into forcing and time-warp terms, growing only linearly in sequence length, not exponentially as in generic switched systems.
- **Computable Certificates:** All stability and robustness constants can be extracted directly from network weights, enabling verification of a trained model at deployment time.

We further extend to general input-dependent $A(u_t)$ via common quadratic Lyapunov functions with efficient Linear Matrix Inequality (LMI) verification. Together, these contributions establish a bridge between classical control theory and modern selective SSM architectures, providing tools for certifying model behavior prior to deployment.

2. Related Work

2.1. State Space Models for Sequence Modeling

The application of state space models to deep learning sequence modeling has a rich history. HiPPO (Gu et al., 2020) introduced the idea of using structured SSM initializations derived from optimal polynomial projections, providing interpretable long-range memory. S4 (Gu et al., 2022) achieved breakthrough performance on the Long Range Arena benchmark by exploiting diagonal-plus-low-rank structure for efficient computation. S5 (Smith et al., 2023) simplified the architecture to single MIMO SSMs with parallel scan implementations. DSS (Gupta et al., 2022) and related work explored diagonal SSM parameterizations.

The gap between SSMs and Transformers for language modeling was addressed by H3 (Fu et al., 2023), which identified key capabilities (recall, comparison) where SSMs underperformed and proposed hybrid architectures. Hyena (Poli et al., 2023) provided an alternative subquadratic operator via long convolutions with data-controlled gating.

Mamba (Gu & Dao, 2024) unified these threads by introducing input-dependent “selective” parameters while retaining efficient parallel scan implementations. The key innovation is making B , C , and the discretization step Δ input-dependent, enabling content-based routing. Mamba-2 (Dao & Gu, 2024) further refined the architecture with structured state space duality. Extensions to vision (Zhu et al., 2024), graphs, and other modalities have followed rapidly.

2.2. Stability of Switched and LPV Systems

Switched linear systems, where dynamics switch among a finite set of modes, and LPV systems, where dynamics vary continuously with parameters, have been extensively studied in control theory (Liberzon, 2003; Shamma & Athans, 1991). A fundamental result is that stability of individual modes does not imply stability of the switched system; counterexamples show that stable matrices can produce unbounded trajectories under fast switching.

Common Quadratic Lyapunov Functions (CQLFs) provide a clean sufficient condition: if a single positive-definite matrix P satisfies $A^\top P A \preceq \beta P$ for all modes, then the switched system is stable (Boyd et al., 1994). For polytopic systems where $A \in \text{conv}\{A_1, \dots, A_K\}$, this reduces to checking vertex LMIs (Gahinet et al., 1996).

The connection between deep learning and control theory has been explored through neural ODEs (Chen et al., 2018), implicit layers (El Ghaoui et al., 2021), and recurrent network stability (Miller & Hardt, 2019). Closer to our setting, StableSSM (Wang & Li, 2024) studies stable reparameterizations of (time-invariant) SSMs, and Halloran et al. (2024) report Lyapunov-stability properties of Mamba training dynamics. Our work differs in providing quantitative BIBO and robustness certificates for the selective scan itself, with weight-extractable constants and an explicit treatment of the time-warp mechanism induced by input-dependent discretization.

2.3. Robustness and Lipschitz Analysis

Lipschitz bounds for neural networks have received significant attention for understanding generalization (Bartlett et al., 2017) and adversarial robustness (Szegedy et al., 2014). For feedforward networks, spectral norm products provide upper bounds; for recurrent networks, the analysis is more delicate due to temporal feedback (Zhang et al., 2018).

Incremental stability and input-to-state stability (ISS) (Sonntag, 2008) provide frameworks for analyzing robustness of dynamical systems. Exponential integrators (Hochbruck & Ostermann, 2010) connect to our variable-step exponential scan interpretation. Learning and identification of stable linear systems (Hardt et al., 2018) provides complementary

perspectives. Our work adapts these tools to the selective SSM setting, providing explicit robustness certificates with computable constants.

3. Main Results

3.1. Problem Setup

We consider the general selective SSM recursion:

$$h_{t+1} = A(u_t)h_t + B(u_t)u_t, \quad (1)$$

$$y_t = C(u_t)h_t + D(u_t)u_t, \quad (2)$$

where $u_t \in \mathbb{R}^m$ is the input at step t , $h_t \in \mathbb{R}^n$ is the hidden state, $y_t \in \mathbb{R}^p$ is the output, and $A(\cdot), B(\cdot), C(\cdot), D(\cdot)$ are input-dependent parameter maps.

The most common selective SSM parameterization, used in Mamba (Gu & Dao, 2024), employs:

$$A_t = e^{\Delta_t A}, \quad \Delta_t = \Delta(u_t) \geq 0, \quad (3)$$

where $A \in \mathbb{R}^{n \times n}$ is a fixed matrix and Δ_t is an input-dependent step size. With exact zero-order hold (ZOH) discretization:

$$h_{t+1} = e^{\Delta_t A} h_t + \int_0^{\Delta_t} e^{(\Delta_t-s)A} ds \cdot B(u_t)u_t. \quad (4)$$

Define accumulated time $\tau_0 = 0$ and $\tau_{t+1} = \tau_t + \Delta_t$. We use $\|\cdot\|$ for a vector norm and its induced operator norm; for symmetric matrices, $X \preceq Y$ denotes Löwner order. Table 3 in the appendix summarizes our notation.

The key insight enabling our analysis is that the discrete selective scan exactly samples a continuous-time forced LTI system.

Lemma 3.1 (Scan-as-sampling Identity). *Define piecewise-constant forcing $w(\sigma) := B(u_t)u_t$ for $\sigma \in [\tau_t, \tau_{t+1})$. Let $h(\sigma)$ solve the continuous-time system*

$$\dot{h}(\sigma) = Ah(\sigma) + w(\sigma), \quad h(0) = h_0. \quad (5)$$

Then the discrete recursion (4) produces $h_t = h(\tau_t)$ for all t .

Proof. Over $\sigma \in [\tau_t, \tau_{t+1})$, forcing is constant. Variation of constants gives $h(\tau_{t+1}) = e^{\Delta_t A} h(\tau_t) + \int_0^{\Delta_t} e^{(\Delta_t-s)A} ds \cdot w(\tau_t)$, matching (4). $\square$

This lemma transforms the analysis of a discrete switched system into analysis of a continuous-time LTI system with piecewise-constant forcing, and it is a much more tractable problem.

3.2. Assumptions

Assumption 3.2 (Fixed- A Scan with Bounded/Lipschitz Maps). Fix an input class $\mathcal{U} = \{u : \|u\| \leq \bar{u}\}$. Assume:

- (A1) **Stable continuous-time core:** A is Hurwitz and $\|e^{tA}\| \leq Me^{-\alpha t}$ for all $t \geq 0$.
- (A2) **Bounded/Lipschitz injection:** $\sup_{u \in \mathcal{U}} \|B(u)\| \leq \bar{B}$ and $\|B(u) - B(\tilde{u})\| \leq L_B \|u - \tilde{u}\|$.
- (A3) **Bounded/Lipschitz readout:** Similarly for C, D with constants $\bar{C}, \bar{D}, L_C, L_D$.
- (A4) **Lipschitz step size:** $\Delta(\cdot)$ is L_Δ -Lipschitz on $\mathcal{U}$.

Define the following derived constants (see Table 4 in the appendix):

$$\bar{w} := \bar{B}\bar{u}, \quad L_w := \bar{B} + L_B\bar{u}, \quad (6)$$

$$\bar{h} := M\|h_0\| + \frac{M}{\alpha}\bar{w}, \quad (7)$$

$$K_{\text{force}} := \bar{C}\frac{M}{\alpha}L_w + L_C\bar{h} + \bar{D} + L_D\bar{u}, \quad (8)$$

$$K_{\text{warp}} := M^2\bar{C}(\|A\|\bar{h} + \bar{w})L_\Delta. \quad (9)$$

With these definitions in place, we are ready to state our main result. The key insight is that the fixed- A exponential parameterization (3), combined with the scan-as-sampling identity (Lemma 3.1), allows us to leverage continuous-time stability theory rather than discrete switched-system analysis. This yields bounds whose sensitivity grows only linearly in sequence length.

3.3. Main Theorem

Theorem 3.3 (BIBO Stability + Output Robustness). *Under Assumption 3.2, let $u_{0:T-1}$ and $\tilde{u}_{0:T-1}$ be two input sequences in $\mathcal{U}$, generating trajectories via (4) from the same initial state $h_0 = \bar{h}_0$, with step sizes $\Delta_t = \Delta(u_t)$ and $\tilde{\Delta}_t = \Delta(\tilde{u}_t)$. Let $\delta := \max_{0 \leq k < T} \|u_k - \tilde{u}_k\|$.*

(i) **BIBO Stability.** *For all $t \leq T$:*

$$\|h_t\| \leq Me^{-\alpha\tau_t} \|h_0\| + \frac{M}{\alpha}\bar{w}, \quad \sup_{t \leq T} \|h_t\| \leq \bar{h}. \quad (10)$$

(ii) **Two-Term Output Robustness.** *For every $0 \leq t \leq T-1$:*

$$\|y_t - \tilde{y}_t\| \leq (K_{\text{force}} + t \cdot K_{\text{warp}})\delta, \quad (11)$$

and therefore $\max_{0 \leq t \leq T-1} \|y_t - \tilde{y}_t\| \leq (K_{\text{force}} + T \cdot K_{\text{warp}})\delta$.

Proof Sketch. Part (i) follows from Lemma 3.1 and standard continuous-time BIBO bounds. For (ii), the two forcings

are piecewise constant on *different* time grids (τ_t vs. $\tilde{\tau}_t$), so a direct forcing-mismatch bound is unavailable. We instead introduce an auxiliary trajectory $\hat{h}$ driven by the *perturbed* inputs on the *unperturbed* grid: the mismatch $h \rightarrow \hat{h}$ is a same-grid forcing perturbation (continuous-time ISS), while $\hat{h} \rightarrow \tilde{h}$ is a pure step-size perturbation controlled by a discrete comparison lemma. See Section B for the complete proof. $\square$

Remark 3.4. The bound (11) grows only linearly in t through the $t \cdot K_{\text{warp}}$ term. This contrasts with generic switched systems, whose sensitivity bounds typically grow exponentially. The linear growth is a direct consequence of the fixed- A structure and the continuous-time interpretation. If, in addition, the step-size map is bounded below on $\mathcal{U}$ by some $\Delta_{\min} > 0$, the time-warp term sharpens to the horizon-uniform constant $M^2 \bar{C}(\|A\| \bar{h} + \bar{w}) L_{\Delta} / (1 - e^{-\alpha \Delta_{\min}})$, so the sensitivity is in fact bounded uniformly in T ; the linear form covers the general case $\Delta(u) \geq 0$.

4. General Stability Guarantees and Practical Verification

The fixed- A analysis covers the most common Mamba parameterization. We now extend to fully input-dependent transitions $A(u_t)$ and develop methods to compute all certificate constants directly from network weights.

4.1. Quadratic Stability for General LPV Systems

When the transition matrix itself depends on input, we require global stability certificates that hold under arbitrary switching.

Definition 4.1 (Common Quadratic Lyapunov Function). A set of matrices $\mathcal{A}$ is quadratically stable if there exist $P \succ 0$ and $\beta \in (0, 1)$ such that $A^\top P A \preceq \beta P$ for all $A \in \mathcal{A}$.

Assumption 4.2 (LPV System with Lipschitz Schedules). On $\mathcal{U} = \{u : \|u\| \leq \bar{u}\}$, assume:

- (B1) **CQLF:** $\exists P \succ 0, \beta \in (0, 1)$ such that $A(u)^\top P A(u) \preceq \beta P$ for all $u \in \mathcal{U}$.
- (B2) **Uniform bounds:** $\sup_{u \in \mathcal{U}} \|B(u)\| \leq \bar{B}$, $\sup_{u \in \mathcal{U}} \|C(u)\| \leq \bar{C}$, $\sup_{u \in \mathcal{U}} \|D(u)\| \leq \bar{D}$.
- (B3) **Lipschitz maps:** $\|A(u) - A(\tilde{u})\| \leq L_A \|u - \tilde{u}\|$, and similarly for B, C, D with constants L_B, L_C, L_D .

Theorem 4.3 (CQLF $\Rightarrow$ BIBO for LPV SSMs). *Under Assumption 4.2(B1)–(B2), define $\alpha = \sqrt{\beta}$ and $M = \kappa_P := \sqrt{\lambda_{\max}(P)/\lambda_{\min}(P)}$. Then the SSM (1)–(2) is BIBO stable: for any input sequence with $\|u_t\| \leq \bar{u}$,*

$$\sup_{t \geq 0} \|h_t\| \leq M \|h_0\| + \frac{M \bar{B}}{1 - \alpha} \bar{u}. \quad (12)$$

Proof. Work in the P -weighted norm $\|h\|_P := \sqrt{h^\top P h}$. The CQLF condition gives $\|A(u)h\|_P \leq \sqrt{\beta} \|h\|_P$, and $\|B(u)u\|_P \leq \sqrt{\lambda_{\max}(P)} \bar{B} \bar{u}$, so $\|h_{t+1}\|_P \leq \sqrt{\beta} \|h_t\|_P + \sqrt{\lambda_{\max}(P)} \bar{B} \bar{u}$. Summing the geometric series in this scalar recursion and converting back via $\sqrt{\lambda_{\min}(P)} \|h\| \leq \|h\|_P \leq \sqrt{\lambda_{\max}(P)} \|h\|$ yields the bound with $M = \kappa_P$ and $\alpha = \sqrt{\beta}$. See Section C for details. Adding the Lipschitz assumption (B3) yields an output robustness certificate analogous to Theorem 3.3. $\square$

Corollary 4.4 (LPV Output Robustness). *Under Assumption 4.2, let $u_{0:T-1}$ and $\tilde{u}_{0:T-1}$ be two input sequences in $\mathcal{U}$ with the same initial state $\hat{h}_0 = h_0$, let $\delta = \max_k \|u_k - \tilde{u}_k\|$, and define*

$$\bar{h} := M \|h_0\| + \frac{M \bar{B}}{1 - \alpha} \bar{u}, \quad (13)$$

$$K_h := \frac{M}{1 - \alpha} (L_A \bar{h} + L_B \bar{u} + \bar{B}), \quad (14)$$

$$K_y := \bar{C} K_h + L_C \bar{h} + L_D \bar{u} + \bar{D}. \quad (15)$$

Then $\max_{t \leq T} \|y_t - \tilde{y}_t\| \leq K_y \delta$, with no dependence on horizon T .

4.2. Polytopic LMI Verification

When $A(u)$ lies in a polytope $\text{conv}\{A_1, \dots, A_K\}$, quadratic stability reduces to a finite set of LMIs.

Proposition 4.5 (Vertex Sufficiency). *If there exist $P \succ 0$ and $\beta \in (0, 1)$ such that $A_i^\top P A_i \preceq \beta P$ for all vertices $i = 1, \dots, K$, then $A^\top P A \preceq \beta P$ for all $A \in \text{conv}\{A_1, \dots, A_K\}$.*

Proof. For $A = \sum_i \xi_i A_i$ with $\xi_i \geq 0$ and $\sum_i \xi_i = 1$, the quadratic form $A^\top P A$ is convex in A , yielding $A^\top P A \preceq \sum_i \xi_i A_i^\top P A_i \preceq \beta P$. $\square$

This enables efficient certification via semidefinite programming. Algorithm 1 uses bisection to find the tightest contraction rate β .

4.3. Computing Constants from Network Weights

We now show how to extract the certificate constants ($M, \alpha, \bar{B}, L_B, \dots$) from the weights of a selective SSM. Combined with Theorem 3.3, this yields deployment-time robustness guarantees.

For the common case of diagonal $A = -\text{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_i > 0$, the semigroup constants are $M = 1$ and $\alpha = \min_i \lambda_i$. For general Hurwitz A , solving the Lyapunov equation $A^\top P + P A = -I$ yields $M = \kappa_P := \sqrt{\lambda_{\max}(P)/\lambda_{\min}(P)}$ and $\alpha = 1/(2\lambda_{\max}(P))$, where P here denotes the continuous-time Lyapunov solution.

Algorithm 1 Polytopic CQLF Certification

Input: Vertices $\{A_i\}_{i=1}^K$, tolerance ϵ
 $\beta_{\text{lo}} \leftarrow 0, \beta_{\text{hi}} \leftarrow 1$
while $\beta_{\text{hi}} - \beta_{\text{lo}} > \epsilon$ **do**
 $\beta \leftarrow (\beta_{\text{lo}} + \beta_{\text{hi}})/2$
Solve SDP: find $P \succ 0$ s.t. $A_i^\top P A_i \preceq \beta P, \forall i$
if feasible **then**
 $\beta_{\text{hi}} \leftarrow \beta$; store P
else
 $\beta_{\text{lo}} \leftarrow \beta$
end if
end while
Output: (P, β_{hi}) if feasible; INFEASIBLE otherwise

The remaining constants depend on the specific parameterization of the input-dependent maps. The following result covers a common gated parameterization; Remark 4.7 treats the official Mamba parameterization (softplus Δ , linear B, C), which the extraction pipeline also handles (see Section 5).

Proposition 4.6 (Lipschitz Constants for Gated Selective-SSM Parameterizations). *Consider the gated parameterizations: step-size map $\Delta(u) = \Delta_{\min} + \Delta_r \sigma(W_\Delta u + b_\Delta)$ where σ is the sigmoid and $\Delta_r := \Delta_{\max} - \Delta_{\min}$, and gated injection $B(u) = \text{diag}(g_B(u))W_B$ with $g_B(u) \in [0, 1]^n$ and g_B being L_{g_B} -Lipschitz. Then the following bounds hold:*

$$L_\Delta \leq \frac{1}{4} \Delta_r \|W_\Delta\|, \quad (16)$$

$$\bar{B} \leq \|W_B\|, \quad L_B \leq L_{g_B} \|W_B\|. \quad (17)$$

Analogous bounds hold for $C(u)$ and $D(u)$ with the same gated structure.

Proof. For (16), the sigmoid satisfies $\|\sigma'\|_\infty = 1/4$, so the chain rule gives $\|\nabla_u \Delta\| \leq \frac{\Delta_r}{4} \|W_\Delta\|$. For (17), since $\|g_B(u)\|_\infty \leq 1$, we have $\|B(u)\| \leq \|W_B\|$. The Lipschitz bound follows from $\|\text{diag}(v)W_B\| \leq \|v\|_\infty \|W_B\| \leq \|v\|_2 \|W_B\|$ applied to $v = g_B(u) - g_B(\bar{u})$. See Section D for details. $\square$

Remark 4.7. The official Mamba implementation uses $\Delta(u) = \text{softplus}(W_\Delta u + b_\Delta)$ (unbounded above) and linear maps $B(u) = W_B u$, $C(u) = W_C u$ rather than $[0, 1]$ -gated forms. Our certificates still apply on a bounded input class: softplus is 1-Lipschitz, so $L_\Delta \leq \|W_\Delta\|$ (no upper bound on Δ is needed by Theorem 3.3); and on $\mathcal{U}$ the linear maps satisfy $\bar{B} \leq \|W_B\| \bar{u}$ and $L_B \leq \|W_B\|$, with the input entering the forcing bilinearly through Lemma B.4. The analysis covers a single selective-scan layer; composing certificates across stacked Mamba blocks with nonlinear mixing is an open problem.

Algorithm 2 Certificate Extraction for Fixed- A Scans

Input: Model with fixed A , parameter networks for B, C, D, Δ ; input bound $\bar{u}$; horizon T
Compute (M, α) from eigenstructure of A
Compute $(\bar{B}, L_B), (\bar{C}, L_C), (\bar{D}, L_D), L_\Delta$ via Proposition 4.6
Compute derived constants:
 $\bar{w} \leftarrow \bar{B}\bar{u}, \quad L_w \leftarrow \bar{B} + L_B \bar{u}$
 $\bar{h} \leftarrow M \|h_0\| + \frac{M}{\alpha} \bar{w}$
 $K_{\text{force}} \leftarrow \bar{C} \frac{M}{\alpha} L_w + L_C \bar{h} + \bar{D} + L_D \bar{u}$
 $K_{\text{warp}} \leftarrow M^2 \bar{C} (\|A\| \bar{h} + \bar{w}) L_\Delta$
Output: Certificate $\max_{t \leq T} \|y_t - \tilde{y}_t\| \leq (K_{\text{force}} + T K_{\text{warp}}) \delta$

Algorithm 2 summarizes the complete extraction procedure. The extracted constants are upper bounds derived from worst-case analysis; actual robustness may be considerably better. Data-dependent refinements that exploit the empirical distribution of u_t are an interesting direction for future work.

5. Experiments

We empirically validate our theoretical framework through systematic experiments on synthetic selective SSMs. Our experiments are organized around four objectives: (1) validating the BIBO stability and two-term robustness decomposition of Theorem 3.3, (2) ablating forcing versus time-warp contributions, (3) verifying LPV certificates via CQLF, and (4) demonstrating the scan-as-sampling identity.

All experiments in Figures 2–3 and Tables 1–2 use a *single* synthetic selective SSM (seed 42): diagonal $A = -\text{diag}(\lambda_1, \dots, \lambda_n)$ with λ_i equally spaced on $[\alpha, \alpha + 1]$, $n = 16$ states, $m = 4$ inputs, $p = 2$ outputs, step sizes $\Delta(u) = \Delta_{\min} + \Delta_r \sigma(W_\Delta u + b_\Delta)$ with $[\Delta_{\min}, \Delta_{\max}] = [0.1, 1.0]$, sigmoid-gated $B(u) = \text{diag}(g_B(u))W_B$ and $C(u) = W_C \text{diag}(g_C(u))$, and a constant feedthrough D ; weight entries are Gaussian with scales $(0.5, 0.2, 0.5, 0.2, 0.5, 0.05)$ for $(W_\Delta, W_B, W_{g_B}, W_C, W_{g_C}, D)$. Inputs are sampled with $\|u_t\| = 0.9\bar{u}$, $\bar{u} = 1$, and $h_0 = 0$. Certificate constants use the spectral-norm bounds of Proposition 4.6. The configuration and extracted constants are summarized in Table 5 (Appendix); Table 1 additionally sweeps n, α , and the step-size range.

5.1. Stability and Robustness Validation

Figure 2 validates Theorem 3.3 across six complementary analyses. Figure 2(a) shows that state trajectories $\|h_t\|$ remain bounded under varying inputs: the actual state norm stays well within the certificate region delimited by the

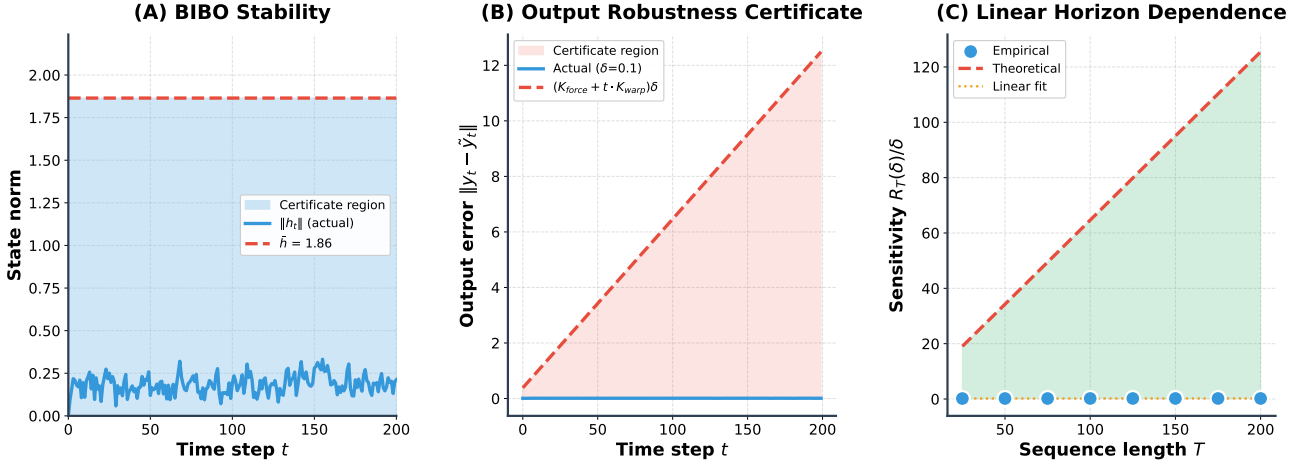


Figure 2. **Theorem 3.3 Validation.** (a) Bounded state trajectories. (b) Output error within the certificate region; the envelope’s intercept is the forcing term and its slope the warp term. (c) Horizon dependence: linear certificate vs. flat measured sensitivity.

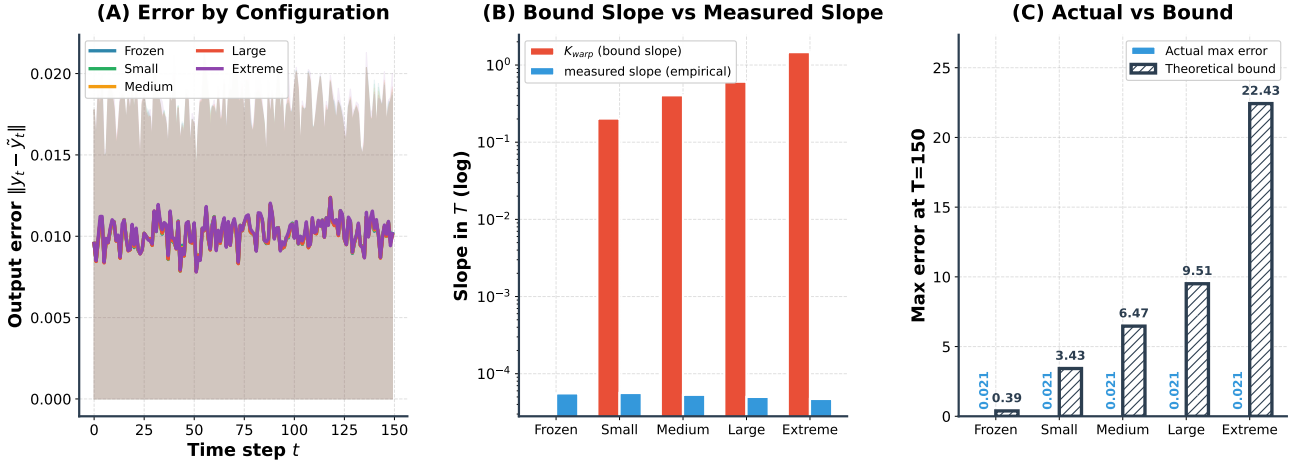


Figure 3. **Forcing vs. Time-Warp Ablation.** (a) Error trajectories under varying step-size ranges. (b) Certificate slope K_{warp} vs. measured sensitivity slope (log scale). (c) Bound vs. actual comparison.

uniform bound $\bar{h} = 1.86$; the time-varying BIBO bound $Me^{-\alpha\tau t}\|h_0\| + \frac{M}{\alpha}\bar{w}$ coincides with it because $h_0 = 0$.

Figure 2(b) demonstrates that the output error $\|y_t - \tilde{y}_t\|$ under perturbation $\delta = 0.1$ remains bounded by the theoretical certificate $(K_{\text{force}} + t \cdot K_{\text{warp}})\delta$. The mean error over 30 random perturbations lies entirely within the certificate region (shaded), confirming the validity of our two-term bound; the envelope’s intercept is the forcing contribution $K_{\text{force}}\delta$ and its slope the time-warp contribution $K_{\text{warp}}\delta$.

A central prediction of Theorem 3.3 is that the certificate slope $R_T(\delta)/\delta$ grows at most linearly in T , in contrast to the exponential envelopes of generic switched systems. Figure 2(c) shows the certificate line $K_{\text{force}} + TK_{\text{warp}}$ growing linearly, while the measured sensitivity under random perturbations remains essentially flat and far below the bound. No exponential blowup is observed at any horizon; the gap

between measurement and certificate reflects the worst-case nature of the constants, quantified further in Section E.

To demonstrate that our certificate extraction pipeline (Algorithm 2) generalizes across architectures, we sweep three axes: state dimension $n \in \{8, 16, 32, 64\}$, stability margin $\alpha \in \{0.3, 0.5, 1.0, 2.0\}$, and step-size range $[\Delta_{\min}, \Delta_{\max}]$. For each configuration we extract the constants $\bar{B}$, $\bar{C}$, L_Δ from the network weights via Proposition 4.6, compute the certificates $\bar{h}$, K_{force} , K_{warp} from Theorem 3.3, and compare against measured values.

Table 1 reports the results. Three trends emerge: (i) Increasing n raises both $\bar{B}$ and $\bar{C}$, which propagates into larger K_{force} and K_{warp} . (ii) Increasing α tightens $\bar{h}$ (from 3.11 at $\alpha=0.3$ to 0.47 at $\alpha=2.0$) and correspondingly reduces K_{force} ($6.30 \rightarrow 1.10$), confirming that a stronger stability margin yields tighter certificates. (iii) The step-size range

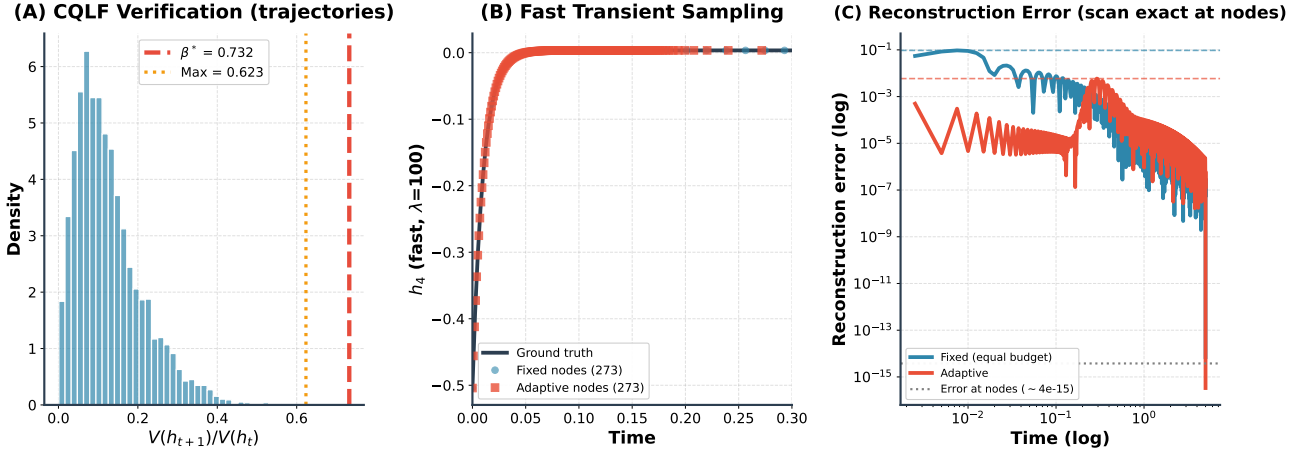


Figure 4. **LPV Certificate and Stiffness Benchmark.** (a) Trajectory contraction ratios $V(h_{t+1})/V(h_t)$ under arbitrary switching vs. the certified β^* from Algorithm 1 (SDP bisection). (b) Node placement in the fast transient of a stiff system. (c) Reconstruction error at a matched node budget (the scan is exact at its nodes).

only affects K_{warp} through L_Δ , while K_{force} stays constant at 3.85 – precisely the two-term decomposition predicted by Theorem 3.3. In all nine configurations, $\max_t \|h_t\| < \bar{h}$, validating the BIBO certificate. The robustness bound $(K_{\text{force}} + TK_{\text{warp}})\delta$ exceeds the measured worst-case output error by two to three orders of magnitude; Section E shows via adversarial perturbation search that this gap reflects genuine conservativeness of the worst-case constants rather than an artifact of random-perturbation evaluation.

To confirm that the pipeline covers the official Mamba parameterization of Remark 4.7 and not only the gated form, we repeat the extraction on a variant with $\Delta(u) = \text{softplus}(w_\Delta^\top u + b_\Delta)$ (unbounded above) and Mamba-style bilinear maps $B(u) = (W_b u)g^\top$, $C(u) = (W_c u)^\top$ ($n = 16$, seed 42). The constants follow from Remark 4.7: $L_\Delta = \|w_\Delta\| = 0.87$, $\bar{B} = 0.93$, $\bar{C} = 1.12$, giving $\bar{h} = 1.86$, $K_{\text{force}} = 6.28$, $K_{\text{warp}} = 3.63$. Both certificates hold ($\max_t \|h_t\| = 0.45 < \bar{h}$; worst measured error 0.038 against a bound of 55.1 at $T = 150$, $\delta = 0.1$). The larger K_{warp} reflects softplus’s unit Lipschitz constant.

5.2. Forcing vs. Time-Warp Ablation

Theorem 3.3 decomposes output sensitivity into a constant forcing term K_{force} and a time-growing warp term $t \cdot K_{\text{warp}}$. We examine how the two certificate terms respond to the step-size parameterization by varying the range ($\Delta_{\text{max}} - \Delta_{\text{min}}$), and test whether the warp mechanism is visible in measured errors.

Figure 3(a) shows error trajectories $\|y_t - \tilde{y}_t\|$ under five configurations of the base model: Frozen ($\Delta_t \equiv 0.55$), Small ($[0.4, 0.7]$), Medium ($[0.25, 0.85]$), Large ($[0.1, 1.0]$, the base range), and Extreme ($[0.05, 1.5]$ with W_Δ scaled $1.5\times$). Under random perturbations the measured errors are

nearly identical across configurations (≈ 0.01 , dominated by same-index forcing and feedthrough sensitivity); the certificates, in contrast, separate sharply.

By construction, K_{force} is constant across configurations (3.85; the B, C, D maps are shared) while $K_{\text{warp}} \propto L_\Delta$ by definition (9): the frozen configuration yields $K_{\text{warp}} = 0$ and the extreme configuration reaches 1.47 (Table 2). Table 2 quantifies the ablation: freezing Δ_t eliminates the time-warp certificate ($K_{\text{warp}} = 0$), while widening the step-size range increases K_{warp} proportionally to L_Δ . We emphasize an key limitation: the measured slopes of the sensitivity in T (last column of Table 2; least-squares fits over $T \in [25, 200]$, 30 random perturbations per horizon) are of order 5×10^{-5} in every configuration — three to four orders of magnitude below K_{warp} , as visualized in Figure 3(b). Random perturbations do not excite the worst-case time-warp mechanism; neither do adversarial ones on this model (extending the attack of Section E across horizons $T \in \{50, \dots, 200\}$ gives an adversarial sensitivity slope of 6.8×10^{-5} in T , against $K_{\text{warp}} = 0.608$). The certificate slope is an upper envelope, not an observed error mechanism.

Figure 3(b) contrasts, on a log scale, the certificate slope K_{warp} against the measured slope for each configuration, making the worst-case/typical-case gap explicit. Figure 3(c) compares actual maximum errors at $T = 150$ (all ≈ 0.021) against the theoretical bounds (0.39 to 22.4). The gap of 1–3 orders of magnitude reflects the worst-case nature of our analysis (see also Section E); tightening these bounds via data-dependent techniques is an important direction for future work.

Table 1. Certificate extraction across model configurations ($T = 150$, $\delta = 0.1$, $\bar{u} = 1.0$; seed 42; “Actual” is the worst case over 20 random perturbations).

n	α	$[\Delta_{\min}, \Delta_{\max}]$	Extracted Constants			Certificates			Validation		
			$\bar{B}$	$\bar{C}$	L_Δ	$\bar{h}$	K_{force}	K_{warp}	$\max \ h_t\ $	Bound	Actual
8	0.5	[0.1, 1.0]	0.68	0.54	0.195	1.35	1.51	0.285	0.26	4.43	0.017
16	0.5	[0.1, 1.0]	0.93	0.84	0.195	1.86	3.85	0.608	0.31	9.51	0.021
32	0.5	[0.1, 1.0]	1.25	0.97	0.195	2.50	6.97	0.945	0.50	14.87	0.018
64	0.5	[0.1, 1.0]	1.81	1.74	0.195	3.61	21.08	2.455	0.69	38.94	0.028
16	0.3	[0.1, 1.0]	0.93	0.84	0.195	3.11	6.30	0.811	0.37	12.79	0.022
16	1.0	[0.1, 1.0]	0.93	0.84	0.195	0.93	2.02	0.456	0.23	7.04	0.021
16	2.0	[0.1, 1.0]	0.93	0.84	0.195	0.47	1.10	0.380	0.17	5.81	0.020
16	0.5	[0.1, 0.5]	0.93	0.84	0.087	1.86	3.85	0.270	0.24	4.44	0.020
16	0.5	[0.1, 2.0]	0.93	0.84	0.412	1.86	3.85	1.284	0.40	19.64	0.022

Table 2. Ablation study: certificate constants vs. measured sensitivity slopes under different step-size configurations.

Configuration	L_Δ	K_{warp}	Measured slope
Frozen ($\Delta_t \equiv 0.55$)	0	0	5.6×10^{-5}
Small ([0.4, 0.7])	0.065	0.203	5.6×10^{-5}
Medium ([0.25, 0.85])	0.130	0.405	5.3×10^{-5}
Large ([0.1, 1.0])	0.195	0.608	5.0×10^{-5}
Extreme ([0.05, 1.5])	0.471	1.469	4.7×10^{-5}

5.3. LPV Certificates and Stiffness Benchmark

For architectures with fully input-dependent $A(u)$, we verify the CQLF-based certificates of Theorem 4.3 on a polytope of $K = 4$ non-symmetric random vertices $\{A_1, \dots, A_4\}$ ($n = 3$, spectral radii $\{0.55, 0.6, 0.5, 0.65\}$, seed 7). This instance is genuinely non-trivial: the identity certificate fails badly ($\max_i \|A_i\|^2 = 3.78 > 1$), so a CQLF must be *found*. Running Algorithm 1 with an interior-point SDP solver (bisection tolerance 10^{-3}) returns a certificate (P, β^*) with $\beta^* = 0.732$, verified a posteriori via the generalized eigenvalue condition $\lambda_{\max}(P^{-1/2} A_i^T P A_i P^{-1/2}) \leq \beta^*$ at every vertex. As a discriminative control, for the classical stable-but-unstable-under-switching pair $A_1 = \begin{pmatrix} 0 & 1.6 \\ 0 & 0 \end{pmatrix}$, $A_2 = A_1^T$ (both nilpotent, but $\rho(A_1 A_2) = 2.56 > 1$), Algorithm 1 correctly reports infeasibility for every $\beta < 1$.

Figure 4(a) shows the distribution of trajectory contraction ratios $V(h_{t+1})/V(h_t)$ over 200 runs of 25 steps with an arbitrary (randomly drawn) convex combination of vertices at each step. All ratios respect the certificate: the observed maximum is 0.623 against the certified $\beta^* = 0.732$.

We next connect variable-step exponential scans to stiff dynamics. Because the exponential scan is *exact* at its own sample points (Lemma 3.1; observed node error $\sim 4 \times 10^{-15}$ on this benchmark), the meaningful comparison is *reconstruction* quality between samples at a matched node budget. On the stiff system $\dot{h} = -\text{diag}(0.1, 1, 10, 100)h + w$

(A) Scan Samples Continuous Trajectory (B) Variable Step Accumulation

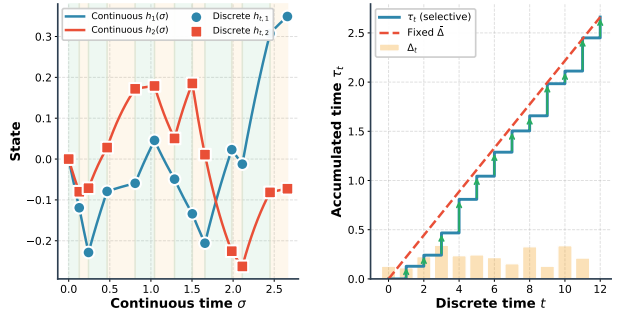


Figure 5. Scan-as-Sampling Identity. (a) Continuous trajectory vs. discrete scan samples at τ_t . (b) Accumulated time τ_t vs. step t , illustrating the time-warp effect.

with constant forcing, we compare a uniform grid against a hand-designed adaptive rule that decreases Δ monotonically with the derivative norm (a saturating sigmoid of $\|\dot{h}\|$ with $\Delta \in [10^{-3}, 0.05]$), both using 273 nodes over $[0, 5]$. Figure 4(b) shows how the adaptive rule concentrates samples in the fast transient ($\lambda = 100$ component). Figure 4(c) reports the reconstruction error (piecewise-linear interpolation between nodes, evaluated against the closed-form solution): the adaptive allocation reduces the *peak* error by $16\times$ ($9.5 \times 10^{-2} \rightarrow 5.9 \times 10^{-3}$), at the cost of slightly larger error in the smooth region where its steps exceed the uniform spacing. This mirrors the resolution-allocation benefit that input-dependent discretization is designed to capture (implemented here, for illustration, with a state-derivative-based rule rather than a learned input-dependent one).

5.4. Scan-as-Sampling Identity

Finally, Figure 5 illustrates the scan-as-sampling identity (Lemma 3.1), which is foundational to our analysis. Figure 5(a) overlays the continuous-time trajectory $h(\sigma)$ with discrete scan samples h_t at sampling times τ_t . The exact

correspondence confirms that the discrete selective scan precisely samples the continuous-time forced LTI system, validating Lemma 3.1.

Figure 5(b) plots accumulated continuous time τ_t versus discrete step t . Under selective step sizes, τ_t grows nonlinearly, while fixed step sizes produce linear growth. The individual step sizes Δ_t illustrate the input-dependent variation. This visualization clarifies the “time-warp” effect: perturbations to input affect not only the forcing but also the effective sampling times, leading to the $t \cdot K_{\text{warp}}$ term in our robustness bound.

5.5. Limitations

Our analysis focuses on a single selective state-space recurrence. It does not provide an end-to-end certificate for a complete Mamba network. The time-warp argument assumes a scalar or channel-shared step size. It also assumes an exact zero-order-hold discretization. Practical Mamba models use channel-dependent step sizes and several additional components. These include gating, convolution, projections, residual connections, normalization, and stacked nonlinear blocks. Our certificates also require known bounds on the inputs and initial states. The LPV analysis further requires a prescribed polytopic enclosure of the admissible system matrices. Our experiments are limited to low-dimensional synthetic systems. The resulting worst-case bounds can therefore be conservative. Future work should cover implemented selective scans and complete Mamba blocks. It should also evaluate certificate tightness and scalability on trained models and real-world tasks.

6. Conclusion

This work addressed the problem of providing formal stability and robustness guarantees for selective State Space Models, whose input-dependent parameterization lies beyond classical LTI analysis. We showed that the fixed- A exponential scan is not an arbitrary switched system but an exact sampling of a continuous-time LTI system on an input-warped time grid. This identity yields BIBO stability and an output-sensitivity certificate that decomposes into a forcing term K_{force} and a time-warp term $t \cdot K_{\text{warp}}$, growing only linearly in sequence length. Beyond this parameterization, the CQLF extension covers general LPV transitions, and all certificate constants are computable directly from trained weights. Stability verification therefore becomes a deployment-time audit rather than a post hoc observation, with the step-size Lipschitz constant identified as the design variable governing long-horizon sensitivity. Future work includes extending the analysis to multi-layer Mamba compositions, certifying pretrained checkpoints, and deriving tighter data-dependent bounds beyond worst-case constants.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, and more specifically, the theoretical understanding of selective State Space Models such as Mamba. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Supplementary Tables

Table 3. Summary of notation.

SYMBOL	MEANING
$u_t \in \mathbb{R}^m$	INPUT AT STEP t
$h_t \in \mathbb{R}^n$	HIDDEN STATE
$y_t \in \mathbb{R}^p$	OUTPUT
$\Delta_t = \Delta(u_t)$	INPUT-DEPENDENT STEP SIZE
$\tau_t = \sum_{k=0}^{t-1} \Delta_k$	ACCUMULATED CONTINUOUS TIME
(M, α)	SEMGROUP CONSTANTS
$\bar{B}, \bar{C}, \bar{D}$	UNIFORM BOUNDS ON MAPS
L_Δ, L_B, L_C, L_D	LIPSCHITZ CONSTANTS
$K_{\text{FORCE}}, K_{\text{WARP}}$	CERTIFICATE CONSTANTS
$P \succ 0, \beta \in (0, 1)$	CQLF PARAMETERS

Table 4. Certificate constants and their roles.

Quantity	Role in Certificate
$\bar{w}, L_w$	Bounds on effective forcing $u \mapsto B(u)u$
$\bar{h}$	Uniform state bound from BIBO stability
K_{force}	Sensitivity from forcing mismatch and readout nonlinearity
K_{warp}	Time-warp slope: sensitivity from step-size changes

Table 5. Experimental configuration and certificate constants extracted from the base selective SSM ($n = 16$ hidden states, seed 42). These values are identical to the $n = 16, \alpha = 0.5, [0.1, 1.0]$ row of Table 1 and to the model used in Figures 2 and 3.

Parameter	Symbol	Value
State / input / output dims	n, m, p	16, 4, 2
Random seed	—	42
Input bound	$\bar{u}$	1.0
Semigroup decay	α	0.5
Semigroup constant	M	1.0
Step size range	$[\Delta_{\min}, \Delta_{\max}]$	$[0.1, 1.0]$
Injection bound	$\bar{B}$	0.93
Readout bound	$\bar{C}$	0.84
Step-size Lipschitz	L_Δ	0.195
Forcing bound	$\bar{w}$	0.93
Forcing Lipschitz	L_w	1.57
State bound	$\bar{h}$	1.86
Forcing certificate	K_{force}	3.85
Time-warp certificate	K_{warp}	0.608

B. Proof of Theorem 3.3

Lemma B.1 (Continuous-time BIBO bound). *Assume $\|e^{tA}\| \leq Me^{-\alpha t}$ and $\sup_{\sigma \geq 0} \|w(\sigma)\| \leq \bar{w}$. Then the solution of $\dot{h} = Ah + w$, $h(0) = h_0$ satisfies for all $\sigma \geq 0$:*

$$\|h(\sigma)\| \leq Me^{-\alpha\sigma} \|h_0\| + \frac{M}{\alpha} \bar{w}.$$

Proof. Variation of constants: $h(\sigma) = e^{\sigma A} h_0 + \int_0^\sigma e^{(\sigma-s)A} w(s) ds$. Bound the integral by $\bar{w} \int_0^\sigma Me^{-\alpha(\sigma-s)} ds \leq \frac{M}{\alpha} \bar{w}$. $\square$

Lemma B.2 (Continuous-time ISS / forcing mismatch bound). *Let $h, \tilde{h}$ solve $\dot{h} = Ah + w$, $\dot{\tilde{h}} = A\tilde{h} + \tilde{w}$ with the same A and $h(0) = \tilde{h}(0)$. If $\sup_{\sigma} \|w(\sigma) - \tilde{w}(\sigma)\| \leq \delta w$, then for all $\sigma \geq 0$:*

$$\|h(\sigma) - \tilde{h}(\sigma)\| \leq \frac{M}{\alpha} \delta w.$$

Proof. $\delta h = h - \tilde{h}$ solves $\dot{\delta h} = A\delta h + \delta w$ with $\delta h(0) = 0$, hence $\delta h(\sigma) = \int_0^\sigma e^{(\sigma-s)A} \delta w(s) ds$ and the same integral bound applies. $\square$

Lemma B.3 (Step-size mismatch for exponential scans). *Let $S(\Delta) := \int_0^\Delta e^{sA} ds$ and let $(\Delta_k)_{k \geq 0}$, $(\tilde{\Delta}_k)_{k \geq 0}$ be two nonnegative step sequences. Given a common forcing sequence (v_k) with $\sup_k \|v_k\| \leq \bar{w}$, define the scans*

$$x_{k+1} = e^{\Delta_k A} x_k + S(\Delta_k) v_k, \quad \tilde{x}_{k+1} = e^{\tilde{\Delta}_k A} \tilde{x}_k + S(\tilde{\Delta}_k) v_k,$$

with $x_0 = \tilde{x}_0$. If $\|e^{sA}\| \leq M e^{-\alpha s}$ for all $s \geq 0$ and $\sup_k \|\tilde{x}_k\| \leq \bar{h}$, then for all $t \geq 0$:

$$\|x_t - \tilde{x}_t\| \leq t M^2 (\|A\| \bar{h} + \bar{w}) \max_{k < t} |\Delta_k - \tilde{\Delta}_k|.$$

If moreover $\Delta_k \geq \Delta_{\min} > 0$ for all k , the factor $t M^2$ improves to $M^2 / (1 - e^{-\alpha \Delta_{\min}})$, uniformly in t .

Proof. Let $e_k := x_k - \tilde{x}_k$, so $e_0 = 0$ and

$$e_{k+1} = e^{\Delta_k A} e_k + (e^{\Delta_k A} - e^{\tilde{\Delta}_k A}) \tilde{x}_k + (S(\Delta_k) - S(\tilde{\Delta}_k)) v_k.$$

From $e^{\Delta_k A} - e^{\tilde{\Delta}_k A} = \int_{\tilde{\Delta}_k}^{\Delta_k} A e^{sA} ds$ we get $\|e^{\Delta_k A} - e^{\tilde{\Delta}_k A}\| \leq M \|A\| |\Delta_k - \tilde{\Delta}_k|$, and likewise $\|S(\Delta_k) - S(\tilde{\Delta}_k)\| = \left\| \int_{\tilde{\Delta}_k}^{\Delta_k} e^{sA} ds \right\| \leq M |\Delta_k - \tilde{\Delta}_k|$. Unrolling with the homogeneous transition $e^{(\tau_t - \tau_{k+1})A}$ (norm at most $M e^{-\alpha(\tau_t - \tau_{k+1})} \leq M$) yields

$$\|e_t\| \leq \sum_{k=0}^{t-1} M \cdot M (\|A\| \bar{h} + \bar{w}) |\Delta_k - \tilde{\Delta}_k|,$$

which gives the linear bound. When $\Delta_k \geq \Delta_{\min}$, we have $\tau_t - \tau_{k+1} \geq (t - k - 1) \Delta_{\min}$, so the geometric sum $\sum_{j \geq 0} e^{-\alpha \Delta_{\min} j} = 1 / (1 - e^{-\alpha \Delta_{\min}})$ replaces the factor t . $\square$

Lemma B.4 (Bilinear forcing map is Lipschitz). *Let $B : \mathcal{U} \rightarrow \mathbb{R}^{n \times m}$ satisfy $\sup_{u \in \mathcal{U}} \|B(u)\| \leq \bar{B}$ and $\|B(u) - B(\tilde{u})\| \leq L_B \|u - \tilde{u}\|$, with $\mathcal{U} = \{u : \|u\| \leq \bar{u}\}$. Then $w(u) = B(u)u$ is bounded by $\bar{w} = \bar{B}\bar{u}$ and Lipschitz with $L_w = \bar{B} + L_B \bar{u}$.*

Proof. Bound $\|w(u)\| \leq \bar{B}\bar{u}$. For Lipschitzness, decompose $B(u)u - B(\tilde{u})\tilde{u} = B(u)(u - \tilde{u}) + (B(u) - B(\tilde{u}))\tilde{u}$ and bound each term. $\square$

Lemma B.5 (Output lift bound). *Assume $\sup_{u \in \mathcal{U}} \|C(u)\| \leq \bar{C}$, $\sup_{u \in \mathcal{U}} \|D(u)\| \leq \bar{D}$, and Lipschitz bounds with constants L_C, L_D . If $\|h(\sigma)\| \leq \bar{h}$ for all σ and $\|u\|, \|\tilde{u}\| \leq \bar{u}$, then*

$$\|C(u)h - C(\tilde{u})\tilde{h}\| \leq \bar{C} \|h - \tilde{h}\| + L_C \bar{h} \|u - \tilde{u}\|,$$

$$\|D(u)u - D(\tilde{u})\tilde{u}\| \leq (\bar{D} + L_D \bar{u}) \|u - \tilde{u}\|.$$

Proof. For the C term, add/subtract $C(\tilde{u})h$: $C(u)h - C(\tilde{u})\tilde{h} = (C(u) - C(\tilde{u}))h + C(\tilde{u})(h - \tilde{h})$. For the D term, add/subtract $D(\tilde{u})u$: $D(u)u - D(\tilde{u})\tilde{u} = (D(u) - D(\tilde{u}))u + D(\tilde{u})(u - \tilde{u})$. Apply the bounds. $\square$

Proof of Theorem 3.3. By Lemma 3.1, the discrete scan satisfies $h_t = h(\tau_t)$ where $h(\cdot)$ solves $\dot{h} = Ah + w$ with piecewise-constant $w(\sigma) = B(u_t)u_t$ on $[\tau_t, \tau_{t+1})$. Under Assumption 3.2(A2), Lemma B.4 gives $\sup_\sigma \|w(\sigma)\| \leq \bar{w}$ and the Lipschitz constant L_w . Lemma B.1 yields (10) and thus the state bound $\bar{h}$ in (7). Applied to the perturbed sequence $\tilde{u}$, the same argument gives $\sup_t \|\tilde{h}_t\| \leq \bar{h}$.

For robustness, note first that the perturbed forcing $\tilde{w}$ is piecewise constant on the *perturbed* grid $\tilde{\tau}_t$, which differs from τ_t ; a pointwise bound on $\|w(\sigma) - \tilde{w}(\sigma)\|$ is therefore unavailable (on misaligned slivers the two forcings carry different discrete indices). We introduce an auxiliary trajectory: let $\hat{h}_k$ be the discrete scan driven by the *perturbed* forcing sequence $v_k := B(\tilde{u}_k)\tilde{u}_k$ but with the *unperturbed* step sizes $\Delta_k = \Delta(u_k)$, initialized at $\hat{h}_0 = h_0$. Split

$$\|h_t - \tilde{h}_t\| \leq \|h_t - \hat{h}_t\| + \|\hat{h}_t - \tilde{h}_t\|.$$

Forcing mismatch (same grid). By the variation-of-constants argument of Lemma 3.1 (which only uses that the forcing is constant on each interval), h_t and $\hat{h}_t$ sample two continuous systems with the same A , the same initial state, and forcings that are piecewise constant on the *same* partition $\{\tau_k\}$: at every $\sigma \in [\tau_k, \tau_{k+1})$ the mismatch is $\|B(u_k)u_k - B(\tilde{u}_k)\tilde{u}_k\| \leq L_w\delta$ by Lemma B.4. Lemma B.2 then gives $\|h_t - \hat{h}_t\| \leq \frac{M}{\alpha} L_w\delta$ (both forcings live on $[0, \tau_T)$, which suffices: the bound at $\sigma = \tau_t$ involves the forcing only on $[0, \tau_t]$).

Time warp (same forcing, different steps). The scans $\hat{h}$ and $\tilde{h}$ share the forcing sequence (v_k) , $\sup_k \|v_k\| \leq \bar{w}$, differ only in their step sizes, and $\sup_k \|\tilde{h}_k\| \leq \bar{h}$. Assumption 3.2(A4) gives $|\Delta_k - \tilde{\Delta}_k| = |\Delta(u_k) - \Delta(\tilde{u}_k)| \leq L_\Delta\delta$, so Lemma B.3 yields

$$\|\hat{h}_t - \tilde{h}_t\| \leq t M^2 (\|A\|\bar{h} + \bar{w}) L_\Delta\delta.$$

Output lift. Apply Lemma B.5:

$$\|y_t - \tilde{y}_t\| \leq \bar{C}\|h_t - \tilde{h}_t\| + (L_C\bar{h} + \bar{D} + L_D\bar{u})\|u_t - \tilde{u}_t\|.$$

Combining the three displays and collecting constants gives (11) with K_{force} and K_{warp} as in (8)–(9). When $\inf_{\mathcal{U}} \Delta \geq \Delta_{\min} > 0$, the sharpened form of Lemma B.3 replaces tM^2 by $M^2/(1 - e^{-\alpha\Delta_{\min}})$, proving the horizon-uniform refinement stated in Remark 3.4. $\square$

C. Proofs of Theorem 4.3 and Corollary 4.4

Lemma C.1 (CQLF implies an induced norm contraction). *Under Assumption 4.2(B1), define $\alpha = \sqrt{\beta}$ and $M = \kappa_P$. Then for any schedule $u_t \in \mathcal{U}$, the homogeneous transition satisfies $\|\Phi(t, s)\| \leq M\alpha^{t-s}$.*

Proof. Let $V(h) = h^\top Ph$. Then $V(h_{k+1}) \leq \beta V(h_k)$, so $V(h_t) \leq \beta^{t-s} V(h_s)$. Convert to Euclidean norm with $\lambda_{\min}(P)\|h\|^2 \leq V(h) \leq \lambda_{\max}(P)\|h\|^2$. $\square$

Lemma C.2 (Uniform state bound under CQLF). *Under Assumption 4.2(B1)–(B2), for any input sequence with $\|u_t\| \leq \bar{u}$:*

$$\sup_{t \geq 0} \|h_t\| \leq \bar{h} := M\|h_0\| + \frac{M\bar{B}}{1 - \alpha}\bar{u}.$$

Proof. Unroll $h_t = \Phi(t, 0)h_0 + \sum_{k=0}^{t-1} \Phi(t, k+1)B(u_k)u_k$. Use Lemma C.1 and $\|B(u_k)\| \leq \bar{B}$, then sum the geometric series. $\square$

Proof of Theorem 4.3. Lemma C.1 establishes $\|\Phi(t, s)\| \leq M\alpha^{t-s}$ under assumption (B1). Lemma C.2 then yields the BIBO bound (13) by unrolling the state recursion and summing the resulting geometric series. $\square$

Lemma C.3 (LPV state incremental bound). *Under Assumption 4.2, for two inputs (u_t) and $(\tilde{u}_t)$ in $\mathcal{U}$ with $h_0 = \tilde{h}_0$ and $\delta = \|u - \tilde{u}\|_\infty$:*

$$\|h_t - \tilde{h}_t\| \leq \frac{M}{1 - \alpha} (L_A\bar{h} + L_B\bar{u} + \bar{B})\delta.$$

Proof. Let $\delta h_t = h_t - \tilde{h}_t$. Subtract updates:

$$\delta h_{t+1} = A(u_t)\delta h_t + (A(u_t) - A(\tilde{u}_t))\tilde{h}_t + (B(u_t) - B(\tilde{u}_t))u_t + B(\tilde{u}_t)\delta u_t.$$

Unroll with $\delta h_0 = 0$, bound using Lemma C.1 and the Lipschitz assumptions, then sum the geometric series. $\square$

Lemma C.4 (LPV output incremental bound). *Under Assumption 4.2 and state bound $\bar{h}$: $\|y_t - \tilde{y}_t\| \leq \bar{C}\|h_t - \tilde{h}_t\| + (L_C\bar{h} + L_D\bar{u} + \bar{D})\delta$.*

Proof. Write $y_t - \tilde{y}_t = C(u_t)h_t - C(\tilde{u}_t)\tilde{h}_t + D(u_t)u_t - D(\tilde{u}_t)\tilde{u}_t$. Add/subtract and use bounds. $\square$

Proof of Corollary 4.4. Lemma C.2 gives (13). Lemma C.3 yields (14). Lemma C.4 combined with (14) yields (15). $\square$

Remark C.5 (Practical Computation of L_{g_B}). In Mamba, the gating $g_B(u)$ is typically parameterized as $g_B(u) = \sigma(W_g u + b_g)$ applied elementwise. By Lemma D.1, $L_{g_B} \leq \frac{1}{4}\|W_g\|$. For more complex gating networks, L_{g_B} can be bounded by the product of layer-wise Lipschitz constants.

D. Proof of Proposition 4.6

Throughout this section, $\|\cdot\|$ denotes the Euclidean norm for vectors and the induced operator (spectral) norm for matrices, and $\|\cdot\|_F$ denotes the Frobenius norm.

Lemma D.1 (Sigmoid is 1/4-Lipschitz). *Let $\sigma(z) = \frac{1}{1+e^{-z}}$. Then for all $a, b \in \mathbb{R}$,*

$$|\sigma(a) - \sigma(b)| \leq \frac{1}{4}|a - b|.$$

Moreover, for the elementwise extension (still denoted σ) acting on vectors $z, \tilde{z} \in \mathbb{R}^r$,

$$\|\sigma(z) - \sigma(\tilde{z})\| \leq \frac{1}{4}\|z - \tilde{z}\|.$$

Proof. For the scalar claim, σ is differentiable and $\sigma'(z) = \sigma(z)(1 - \sigma(z)) \in (0, \frac{1}{4}]$ for all $z \in \mathbb{R}$, with maximum $\frac{1}{4}$ attained at $z = 0$. By the mean value theorem, $|\sigma(a) - \sigma(b)| \leq \sup_z |\sigma'(z)| |a - b| \leq \frac{1}{4}|a - b|$.

For the vector claim, apply the scalar bound coordinatewise:

$$\|\sigma(z) - \sigma(\tilde{z})\|^2 = \sum_{i=1}^r |\sigma(z_i) - \sigma(\tilde{z}_i)|^2 \leq \sum_{i=1}^r \left(\frac{1}{4}|z_i - \tilde{z}_i|\right)^2 = \left(\frac{1}{4}\right)^2 \|z - \tilde{z}\|^2,$$

which implies $\|\sigma(z) - \sigma(\tilde{z})\| \leq \frac{1}{4}\|z - \tilde{z}\|$. $\square$

Lemma D.2 (Diagonal gating bounds). *Let $v \in \mathbb{R}^n$ and let W be any real matrix with compatible dimensions. Then:*

$$\begin{aligned} \|\text{diag}(v)W\| &\leq \|v\|_\infty \|W\|, \\ \|\text{diag}(v)W\| &\leq \|v\| \|W\|_F. \end{aligned}$$

Similarly, if $W \in \mathbb{R}^{p \times n}$, then

$$\begin{aligned} \|W \text{diag}(v)\| &\leq \|v\|_\infty \|W\|, \\ \|W \text{diag}(v)\| &\leq \|v\| \|W\|_F. \end{aligned}$$

Proof. For the first inequality, use submultiplicativity and $\|\text{diag}(v)\| = \|v\|_\infty$:

$$\|\text{diag}(v)W\| \leq \|\text{diag}(v)\| \|W\| = \|v\|_\infty \|W\|.$$

For the second inequality, use $\|X\| \leq \|X\|_F$ and compute

$$\|\text{diag}(v)W\|_F^2 = \sum_{i=1}^n v_i^2 \|W_{i,:}\|_2^2 \leq \left(\sum_{i=1}^n v_i^2\right) \left(\sum_{i=1}^n \|W_{i,:}\|_2^2\right) = \|v\|^2 \|W\|_F^2,$$

so $\|\text{diag}(v)W\| \leq \|\text{diag}(v)W\|_F \leq \|v\| \|W\|_F$.

The right-multiplication bounds for $W \text{diag}(v)$ follow by the same argument, noting that $\|W \text{diag}(v)\| \leq \|W\| \|\text{diag}(v)\|$ and $\|W \text{diag}(v)\|_F^2 = \sum_{j=1}^n v_j^2 \|W_{:,j}\|_2^2 \leq \|v\|^2 \|W\|_F^2$. $\square$

With these lemmas in hand, we now prove the two claimed bounds separately.

Step-size map. Let

$$\Delta(u) = \Delta_{\min} + (\Delta_{\max} - \Delta_{\min}) \sigma(W_{\Delta} u + b_{\Delta}),$$

where σ is applied elementwise (the scalar case is a special case). For any $u, \tilde{u}$,

$$\begin{aligned} \|\Delta(u) - \Delta(\tilde{u})\| &= (\Delta_{\max} - \Delta_{\min}) \|\sigma(W_{\Delta} u + b_{\Delta}) - \sigma(W_{\Delta} \tilde{u} + b_{\Delta})\| \\ &\leq (\Delta_{\max} - \Delta_{\min}) \cdot \frac{1}{4} \|W_{\Delta}(u - \tilde{u})\| \quad (\text{by Lemma D.1}) \\ &\leq \frac{1}{4} (\Delta_{\max} - \Delta_{\min}) \|W_{\Delta}\| \|u - \tilde{u}\|. \end{aligned}$$

Hence the Lipschitz constant satisfies

$$L_{\Delta} \leq \frac{1}{4} (\Delta_{\max} - \Delta_{\min}) \|W_{\Delta}\|.$$

Gated injection map. Let $B(u) = \text{diag}(g_B(u))W_B$ with $g_B(u) \in [0, 1]^n$ for all u . Since $\|g_B(u)\|_{\infty} \leq 1$, Lemma D.2 gives

$$\|B(u)\| = \|\text{diag}(g_B(u))W_B\| \leq \|g_B(u)\|_{\infty} \|W_B\| \leq \|W_B\|,$$

so $\bar{B} \leq \|W_B\|$.

For the Lipschitz bound, for any $u, \tilde{u}$,

$$B(u) - B(\tilde{u}) = (\text{diag}(g_B(u)) - \text{diag}(g_B(\tilde{u})))W_B = \text{diag}(g_B(u) - g_B(\tilde{u}))W_B.$$

Applying Lemma D.2 (the ℓ_{∞} -based bound) together with $\|v\|_{\infty} \leq \|v\|_2$ yields

$$\|B(u) - B(\tilde{u})\| \leq \|g_B(u) - g_B(\tilde{u})\|_{\infty} \|W_B\| \leq \|g_B(u) - g_B(\tilde{u})\|_2 \|W_B\| \leq L_{g_B} \|u - \tilde{u}\| \|W_B\|,$$

so

$$L_B \leq L_{g_B} \|W_B\|.$$

(The Frobenius-based bound of Lemma D.2 gives the weaker $L_B \leq L_{g_B} \|W_B\|_F$; since $\|W_B\| \leq \|W_B\|_F$, the spectral form above is tighter and is the one used in our experiments.)

Extension to $C(u)$ and $D(u)$. If $C(u)$ (or $D(u)$) admits the same gated form, e.g. $C(u) = \text{diag}(g_C(u))W_C$ or $C(u) = W_C \text{diag}(g_C(u))$ with $g_C(u) \in [0, 1]^r$ and $\|g_C(u) - g_C(\tilde{u})\| \leq L_{g_C} \|u - \tilde{u}\|$, then the identical argument using Lemma D.2 gives

$$\bar{C} \leq \|W_C\|, \quad L_C \leq L_{g_C} \|W_C\|,$$

and likewise for D with (W_D, g_D, L_{g_D}) . This proves all claims.

E. Certificate Tightness under Adversarial Perturbations

The validation columns of Table 1 evaluate the robustness certificate against *random* perturbations, which may substantially underestimate the true worst case and hence overstate the conservativeness of the bound. To disentangle the two effects, we search for worst-case perturbations directly on the base model ($T = 150$, $\delta = 0.1$): projected gradient ascent over the per-step perturbation ball, with finite-difference gradients, 100 ascent iterations, and 6 restarts (one warm-started at the best of 2,000 random perturbations).

The outcome: the adversarial search improves only marginally on random sampling (0.0247 vs. 0.0237), indicating that random evaluation is not the bottleneck for this model class: the certificate value of 9.51 exceeds the strongest perturbation found by a factor of ~ 385 . Extending the attack across horizons $T \in \{50, 100, 150, 200\}$ yields adversarial sensitivities $\{0.222, 0.240, 0.232, 0.236\}$ — a fitted slope of 6.8×10^{-5} in T , four orders of magnitude below $K_{\text{warp}} = 0.608$ — so even worst-case-seeking perturbations do not excite the time-warp growth on this model. We conclude that the gap in Table 1 predominantly reflects genuine conservativeness of the worst-case constants (M/α products, global Lipschitz bounds, and the triangle-inequality composition), and we view data-dependent tightening — e.g., replacing global Lipschitz constants with local ones along the observed state distribution — as the most promising route to practically tight certificates.